

Understanding CO₂ Emission Drivers Across EU Countries (2018–2021)

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1. Introduction:

The research question: *What factors influence the variation in CO₂ emissions among EU member states?*

This question arises from the notable differences in CO₂ per capita emissions between EU countries and the diversity in climate targets and achievements among member states. Understanding these factors is crucial for informing EU-wide climate policies, especially as the organization strives to meet its ambitious 2050 climate neutrality goals.

2. Literature Review:

A review of prior studies highlights key trends and gaps in the literature;

Bianco, V., Cascetta, F., & Nardini, S. (2024) explores the CO₂ emissions trends in the EU between the years of 1995 and 2019 using a method called Logarithmic Mean Divisia Index (LMDI). Their findings highlight the critical role of technical advancements and climatic conditions in reducing emissions, contrasting with the upward pressure from socio-economic growth.

Padilla, E., & Duro, J. A. (2013) study the CO₂ emissions per capita inequality among the EU states and explores the inequality from the data between the years 1990 and 2009. The study's findings are that energy intensity has contributed to the reduction of inequality and uses GDP per capita to show the disparity between these states. The study emphasizes inequality both within and between regions.

Loures, L., & Ferreira, P. (2019) conducts a qualitative comparative analysis by looking at the relationship between energy consumption and CO₂ emissions in the EU between the years 2010 and 2014. They support the Environmental Kuznets Curve (EKC) hypothesis by demonstrating that economic growth initially increases the emissions, but the peak later drops due to the implementations of cleaner technologies.

Fernández González, P., Landajo, M., & Presno, M. J. (2013) investigate changes in CO₂ emissions across EU states (2001–2008). The study attributes emissions reductions to energy efficiency and fuel mix changes but warns of continued pressure from population and

economic growth, underscoring the importance of integrating sustainability with development.

Axelsson, K., et al. (2024) shift the focus to consumption-based emissions (CBEs), revealing that territorial emissions underrepresent the EU's carbon footprint. They highlight the significant contribution of imported goods and suggest policy measures addressing global trade impacts. This provides an important perspective for addressing global emissions' disparities.

Martínez-Zarzoso, I., et al. (2007) uses the STIRPAT model, to study the impact of population growth on CO₂ emissions between the years 1975 and 1999 among EU member states. The study showcases the significant difference between the member ship dates between the states. The member ship especially shows significance in population and economic pressures. The study tries to relay the importance of regional variability.

Duro, J. A., & Padilla, E. (2013) researches the polarization in CO₂ emissions per capita across EU countries between the years 1990 and 2007, finding a reduction in polarization due to energy efficiency improvements. By assessing within- and between-region disparities, the study demonstrates progress in emissions convergence which facilitates policy consensus across regions.

However, these studies often overlook regional disparities or fail to integrate multiple socio-economic, technical, and political variables in a single analysis. This project seeks to address these gaps. While existing studies explore various aspects of CO₂ emissions in the EU such as inequality, energy intensity, and consumption-based factors, this research seeks to integrate these insights by considering a comprehensive set of explanatory variables and accounting for regional variability. By focusing on a variety of variables that tries to encapsulate all spheres of development, our study aims to offer a holistic approach on the drivers of CO₂ emissions.

We hypothesize that multiple factors influence CO₂ emissions per capita, including GDP per Capita, Industrialization Rate, Urbanization Rate, Educational Attainment, Gender Equality, Corruption Rate, Democracy Index, Climate Change Performance Index (CCPI), EU Join Year, regional grouping (e.g., Western Europe, Former Eastern Bloc), and Renewable Energy Usage. For instance, we expect higher GDP per Capita and Industrialization Rates to be associated with increased CO₂ emissions per capita, while higher Renewable Energy Usage and Gender Equality are expected to reduce emissions.

This study aims to be useful as the EU is a global leader in climate change mitigation, yet its member states exhibit considerable variability in emissions, policies, and progress. This study aims to uncover the underlying factors that contribute to these differences, providing

valuable insights for policymakers. By analyzing a diverse set of socio-economic, technical, and political variables and accounting for regional differences, this research offers a nuanced perspective on emissions variability within a single policy framework. The findings can inform targeted strategies to help the EU collectively achieve its 2050 climate neutrality goal.

3. Method:

For this study we used purposive sampling which is a sampling method in which the researchers intentionally select the study participants based on a specific criterion, in our case our criteria is a state should be an EU member states and the data should be from a specific time frame (National Center for State Courts, n.d.).

The dataset includes observations from 27 EU member states over four years (2018–2021). The primary data source is the World Bank, which provides consistent and complete data on variables such as GDP per Capita (in USD), Urbanization Rate (percentage of total population), and Educational Attainment (percentage of the population with at least a bachelor's degree). Additional variables, such as the Climate Change Performance Index and Democracy Index, are sourced from specialized reports. CO₂ emissions per capita, measured in metric tons of CO₂ excluding LULUCF (Land Use, Land-Use Change, and Forestry), serve as the dependent variable. The dataset is complete, with no missing data points during the specified period.

To analyze the data, we will first conduct a descriptive statistical analysis. This includes generating summary statistics to examine the distribution of variables across EU regions. Scatterplot matrices and correlation analyses will be used to explore potential linear relationships between explanatory variables and CO₂ emissions, as well as to identify multicollinearity or outliers. Based on this analysis, we will select the final explanatory variables to include in the model and determine whether data transformations are necessary.

Following that, we plan to employ multiple linear regression modeling to evaluate the relationship between CO₂ emissions per capita and the explanatory variables. Given that our dataset includes four consecutive years of data for each country, we will use the country as panel data in the model to address potential autocorrelation and improve robustness. We will also conduct assumption checks to ensure the credibility of the model results. Furthermore, certain dummy variables were excluded from the final model due to aliased collinearity with other variables. Thus, we performed separate analyses on these variables to capture their potentially significant information.

4. Result:

Exploratory Data Analysis

Number of Observations: Number of total observations is 108. These data are grouped by regions: Western Europe has 24 observations, Former Eastern Bloc has 44 observations, Northern Europe 16 observations, and Southern Europe 24 observations.

As it can be seen from the summary statistics analysis tables there are notable differences in CO₂ emissions and contributing factors across EU regions. Western Europe has moderate CO₂ emissions per capita (mean of 8.7) with relatively uniform distribution, but its total emissions are highly skewed due to outliers, reflecting higher GDP per capita and low renewable energy use (17.9%). The Former Eastern Bloc has slightly lower per capita emissions (mean of 6.5) but significant variability in total emissions, driven by heavy industrial reliance and moderate renewable energy use (24.6%). Northern Europe exhibits the lowest CO₂ emissions per capita (mean: 6.0 t) and total emissions, attributed to high renewable energy adoption (41.9%) and effective governance. Southern Europe, with the lowest per capita emissions (mean: 4.9 t), faces economic constraints, moderate renewable energy use (17.8%), and lower labor force participation (49.8%). Across regions, Northern Europe leads in governance, renewable energy, and gender inclusivity, while Western Europe and Southern Europe show room for improvement, particularly in renewable energy investments.

Analysis of Scatterplot Matrices

From Figure 3, countries with higher democracy (strong positive correlation, 0.883) scores tend to have lower corruption levels, consistent with the notion that democratic governance reduces corruption. Moreover, higher corruption levels are associated with higher CO₂ emissions per capita (moderate positive correlation, 0.426), suggesting that governance issues may hinder effective climate policy implementation. Finally, a slight trend (weak positive correlation, 0.266) suggests that more democratic countries might have slightly higher CO₂ emissions, likely due to industrial activity and wealth.

From Figure 4, country groups reveal that CO₂ per capita emissions are as follows Northern Europe > Southern Europe > Former Eastern Bloc > Western Europe (outliers like Luxembourg should be remembered). This is likely due to the CO₂ emissions increase and peak associated with higher levels of development. However, there is a geopolitical difference between these states which highly impacts their energy fuel participation, which can be seen in Figure 7. According to it countries with higher renewable energy usage tend to have lower CO₂ emissions, as expected due to reduced reliance on fossil fuels. Following this trend, hydropower is a highly utilized renewable energy source for most Eastern European Bloc countries. The availability of water sources as a fuel source results in a

natural reduction of CO₂ emissions. Moreover, for Southern Europe, Spain generates a large chunk of its energy from windmills (Hauer and Habersack, 2019). On the other hand, for most Western European states the most utilized form of electricity production was Nuclear however current cultural fear and push from green parties has resulted in the shutting down of these facilities in places like Germany Hauer and Habersack, 2019). Going back to Figure 4, countries that joined the EU earlier tend to have higher CO₂ emissions per capita, possibly reflecting their industrial development prior to EU accession.

Figure 5 illustrates the relationships between education, urban population, and CO₂ emissions per capita. A moderate positive correlation (0.409*) exists between education level and CO₂ emissions, indicating that wealthier, industrialized countries with better education systems tend to emit more CO₂ per capita. Similarly, urban population shows a moderate positive correlation (0.431) with CO₂ emissions, reflecting the higher energy consumption and industrialization typical of urbanized nations. In contrast, school enrollment demonstrates a very weak negative correlation (-0.061), suggesting minimal direct impact on CO₂ emissions.

Figure 6 examines the relationships between GDP, labor force participation, industry share, and CO₂ emissions per capita. A moderate positive correlation (0.495) between GDP per capita and CO₂ emissions suggests that wealthier countries with higher industrial output and energy use emit more CO₂ per capita. Similarly, labor force participation shows a moderate positive correlation (0.471), indicating that countries with larger labor markets and industrial economies tend to have higher emissions. In contrast, industry share of GDP exhibits a weak negative correlation (-0.115), suggesting that increased industrial activity does not necessarily lead to higher emissions, possibly due to cleaner production methods in more developed economies.

Figure 7 explores the relationships between and CO₂ emissions per capita. A moderate negative correlation (-0.305) indicates that better CCPI rankings are associated with lower CO₂ emissions, highlighting the effectiveness of robust climate policies. Boxplot analysis further reveals that high-ranked CCPI countries have significantly lower and more consistent CO₂ emissions, while poorly ranked countries show higher and more variable emissions, emphasizing disparities in climate performance.

The exploratory data analysis and scatterplot matrices provide evidence supporting several of our initial hypotheses. The data confirms that factors such as GDP per capita, industrialization, urbanization, and governance significantly influence variations in CO₂ emissions per capita among EU member states. Regions with higher renewable energy adoption, such as Northern Europe, exhibit lower emissions, aligning with our hypothesis about the mitigating effects of renewable energy usage. Similarly, the role of governance is

evident in the strong correlations between democracy scores, corruption levels, and emissions, reinforcing our expectations about governance quality impacting climate policy effectiveness. These findings align with our hypothesis that regional differences and socio-economic variables play a crucial role in explaining emissions variability across the EU. Out of ordinary or unexpected events are easily explained by cultural, political and geological phenomenas.

Panel Data Model

In the initial Fixed Effects Model, the presence of heteroscedasticity was identified as a key issue. To address this, the final model employed a Fixed Effects Model with robust standard errors, ensuring more reliable and robust results. The final model yielded an F-statistic with a p-value less than 0.01, indicating that at least one explanatory variable significantly influences the dependent variable. The analysis results confirm that the significant explanatory variables are consistent: log_GDPperCapita, log_ReneEnergy, and UrbPOP. Specifically:

- Every 1% increase in GDP per capita is associated with a 0.22% increase in CO₂ emission per capita.
- Every 1% increase in Renewable Energy proportion is associated with a 0.74% decrease in CO₂emission per capita.
- Every 1% increase in Urban Population proportion is associated with a 0.08% decrease in CO₂ emission per capita.

While variables such as educational attainment, industry value-added proportion, female labor force participation rate, school enrollment gender parity index, democracy index, and CCPI did not show a statistically significant relationship with CO₂ emissions per capita in this model.

The adjusted R-squared value of the model is 0.4409, meaning that approximately 44.09% of the variation in the dependent variable is explained by the model. This demonstrates a relatively strong explanatory power.

Assumption Test

The Residuals vs. Fitted plot (Figure. x) showed that the residuals were evenly distributed around zero without any clear pattern, indicating that the linearity assumption was largely satisfied. However, the Breusch-Pagan test (BP = 56.655, $p < 0.05$) revealed significant heteroscedasticity, suggesting that the variance of residuals is not constant across the observations.

The Q-Q plot (Figure. x) showed that most residuals aligned closely with the diagonal line, with only slight deviations in the tails, suggesting that the residuals approximately follow a normal distribution. On the other hand, the analysis of multicollinearity revealed relatively high GVIF values for some variables, likely due to the inclusion of numerous dummy variables for factor (Country), which may have introduced overfitting, which has been removed in our final model. Additionally, the Cook's Distance plot (Figure. x) identified a few influential observations, particularly observations 29, 32, and 81, which may impact the model's estimates.

In summary, by addressing the heteroscedasticity issue using robust standard errors, the final Fixed Effects Model ensures more reliable results while retaining its explanatory power. The model satisfactorily meets other assumptions, confirming the credibility of our results.

Difference Test

In the final model, the country group variable was excluded, but scatterplot analysis clearly showed differences in CO₂ emissions per capita across country groups. Therefore, we conducted statistical tests to verify these differences. The Shapiro-Wilk test results indicated that, except for Northern Europe ($p = 0.20$), the distributions for Eastern Block ($p < 0.01$), Southern Europe ($p = 0.02$), and Western Europe ($p < 0.01$) deviate from normality, which was also reflected in previous histograms. Consequently, for multi-group comparisons with non-normal distributions, we referred to the methodology in Schneider-Mayerson et al. (2020) and applied the Kruskal-Wallis test followed by Dunn's post-hoc test.

The Kruskal-Wallis test results showed $df = 3$, $p < 0.01$, indicating significant differences between at least one pair of country groups, with the post-hoc Dunn test showed the following significant differences:

- Eastern Block vs. Southern Europe ($P_{adj} = 0.0494$, 6.5 vs. 4.9);
- Eastern Block vs. Western Europe ($P_{adj} = 0.0118$, 6.5 vs. 8.7);
- Northern Europe vs. Western Europe ($P_{adj} = 0.0050$, 6.0 vs. 8.7);
- Southern Europe vs. Western Europe ($P_{adj} = 0.0020$, 4.9 vs. 8.7).

No significant differences were found between Eastern Block and Northern Europe, or between Southern Europe and Northern Europe.

5. Discussion

In conclusion, the final model used in this study demonstrates robust performance and effectively explains some of the factors influencing CO₂ emission per capita among EU countries between 2018 and 2021.

The analysis reveals that an increase in GDP per capita leads to higher CO₂ emissions. However, as suggested by Cole et al. (1997), the relationship between GDP and CO₂ emissions may not be strictly linear but rather exhibit a U-shaped curve. This indicates that the current model may have limitations in capturing potential non-linear relationships. The proportion of renewable energy usage shows the most significant negative correlation with CO₂ emissions per capita, with the largest coefficient among all variables. This highlights the crucial role of renewable energy in reducing carbon emissions. Surprisingly, the proportion of urban population is negatively correlated with CO₂ emissions per capita. This may be due to the higher energy efficiency resulting from concentrated production and living activities in urban regions.

On the other hand, several variables were found to have no significant effect on CO₂ emissions per capita. Educational attainment, gender equality, and democracy levels showed no significant impact, probably because their effects are masked by other variables in the model. The insignificance of the industry value added proportion may result from the variable reflecting only the proportion of industrial activities within the economy, without accounting for the internal structure of the industrial sector. Notably, the Climate Change Performance Index (CCPI) was not a significant factor. This could be due to a gap between policy formulation and implementation efficiency or differences in baseline CO₂ emissions across countries.

The difference analysis also showed unexpected results. Contrary to our initial hypothesis that Eastern Europe would have higher CO₂ emissions per capita due to political and economic conditions, Western Europe was found to have significantly higher emissions. This outcome can be attributed to Eastern Europe's greater reliance on renewable energy like hydropower, while Western Europe's highest GDP and lowest renewable energy usage contribute to its elevated emissions, which also aligns with regression analysis. These findings revealed that adopting alternative energy sources such as hydropower to balance industrialization with carbon emission reduction is doable yet crucial, offering valuable insights especially for developing countries.

This study also has several limitations. For example, data limitations for some variables restricted the sample period to 2018–2021, resulting in a relatively small sample size. Thus, some conclusions might be constrained. Besides, as observed in Figures 1 and 2, CO₂ emission per capita across countries significantly declined in 2020, likely due to the impacts of the COVID-19 pandemic. The specific mechanisms and long-term implications of this effect call for further investigation. Moreover, the findings of our study raise additional hypotheses regarding factors influencing CO₂ emissions in EU countries, such as the usage

of natural gas and other energy sources, as well as the internal structure of industrial sectors. These aspects warrant further exploration in future research.

Appendix:

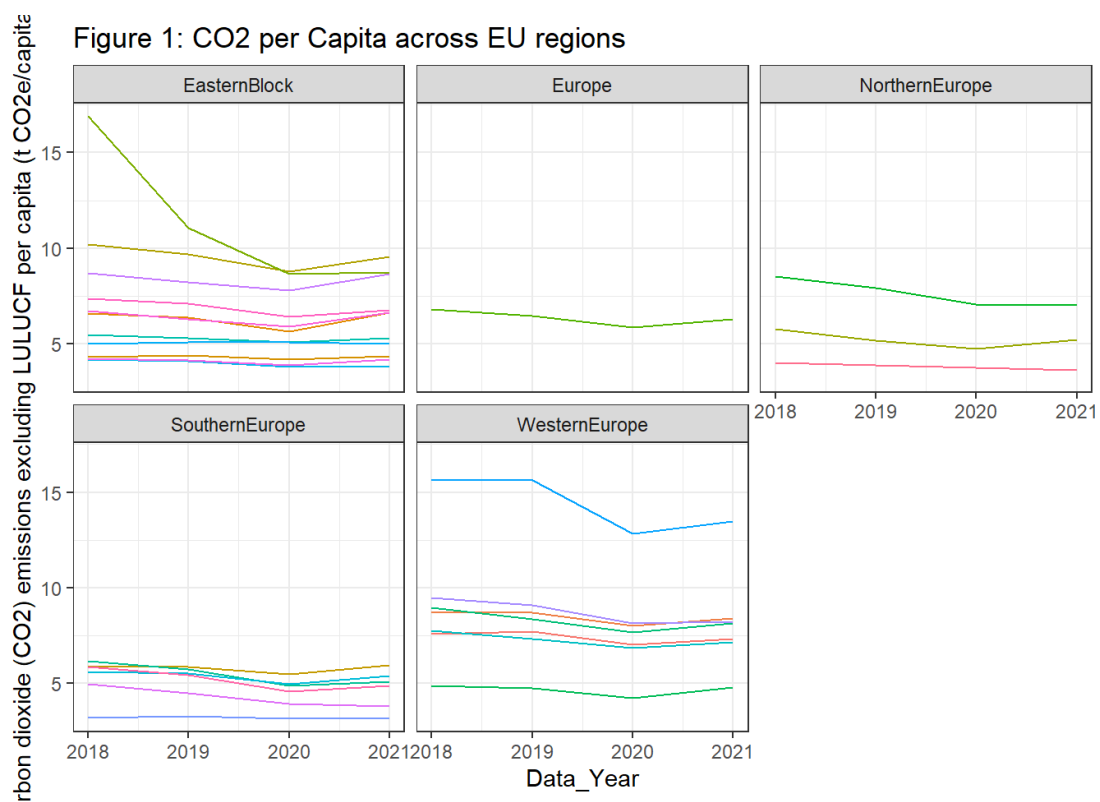


Figure 2: Annual CO2 emissions per Capita across years.

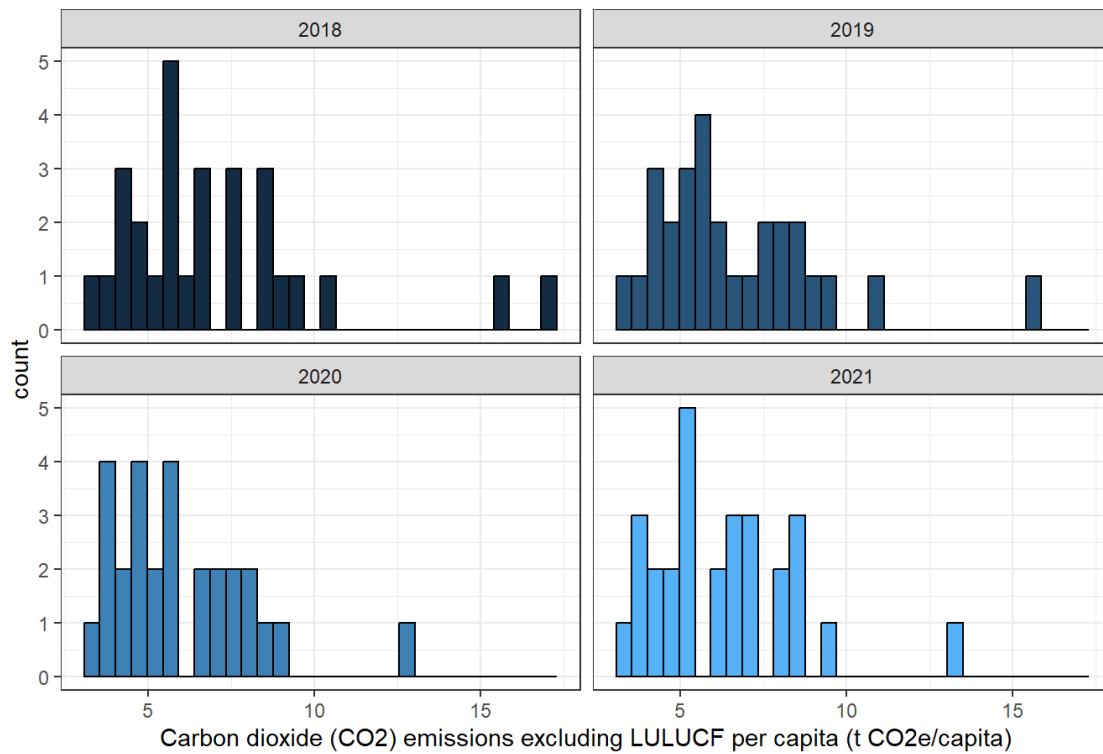


Figure 3: Scatterplot Matrix for EU Comparison

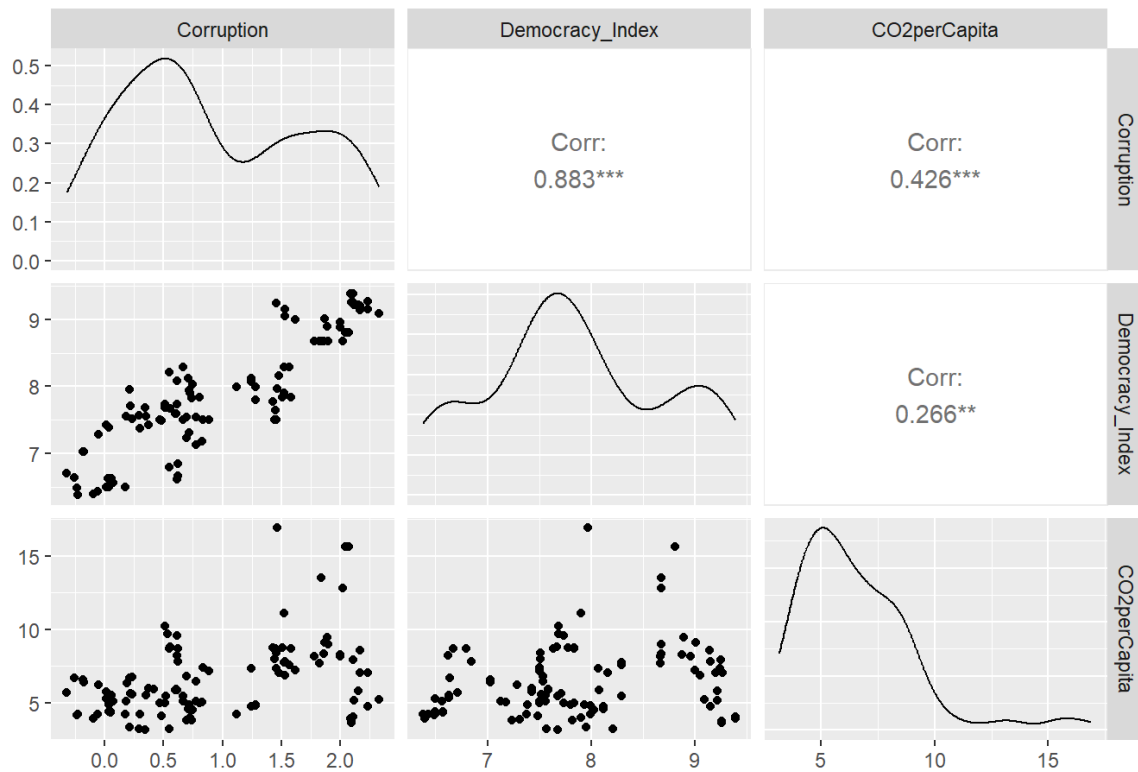


Figure 4: Scatterplot Matrix for EU

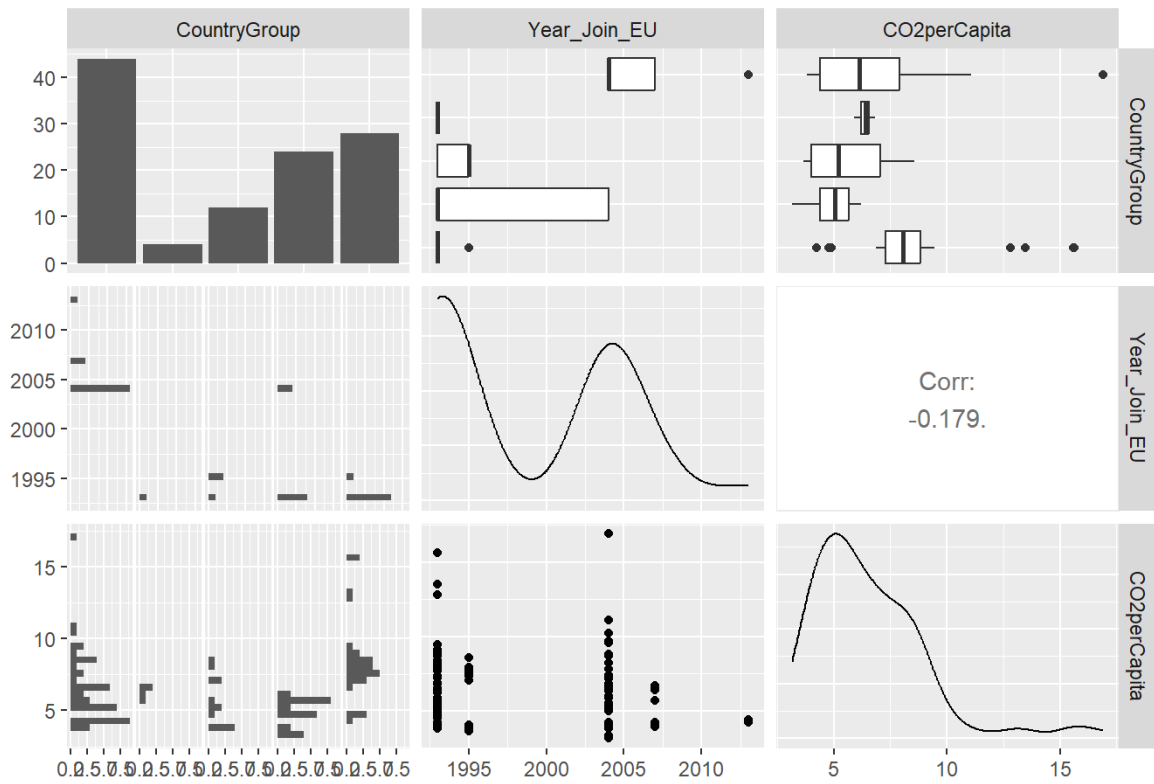


Figure 5: Scatterplot Matrix for EU

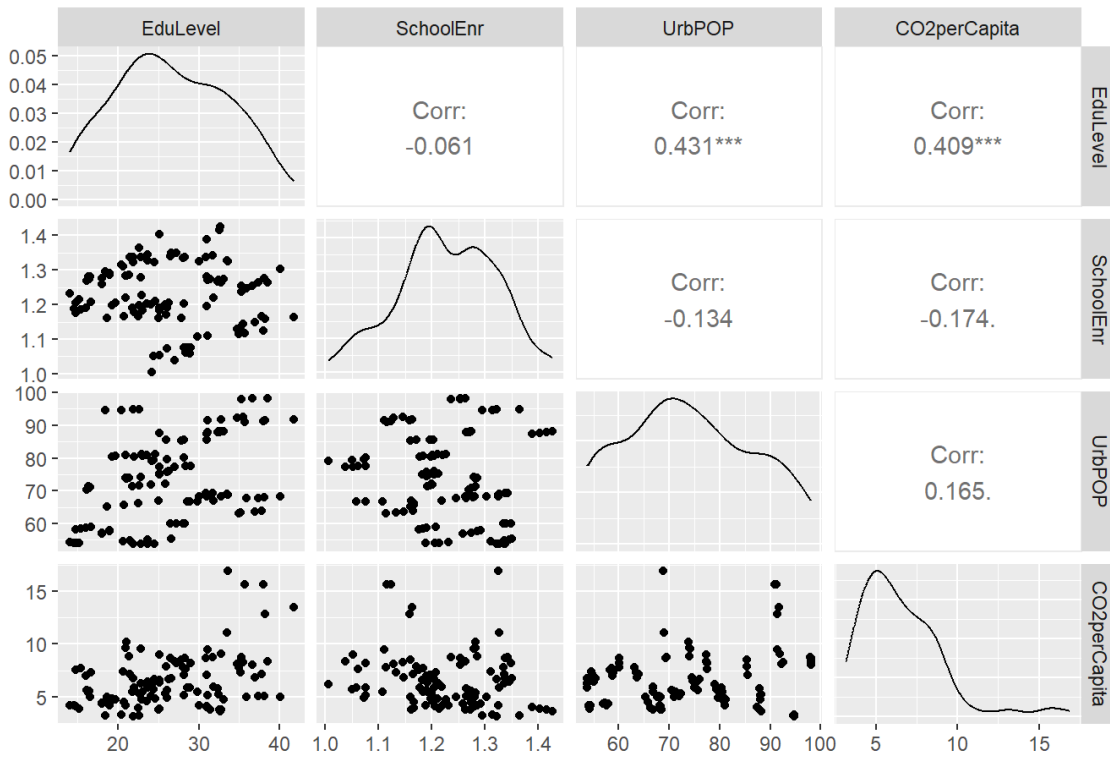


Figure 6: Scatterplot Matrix for EU

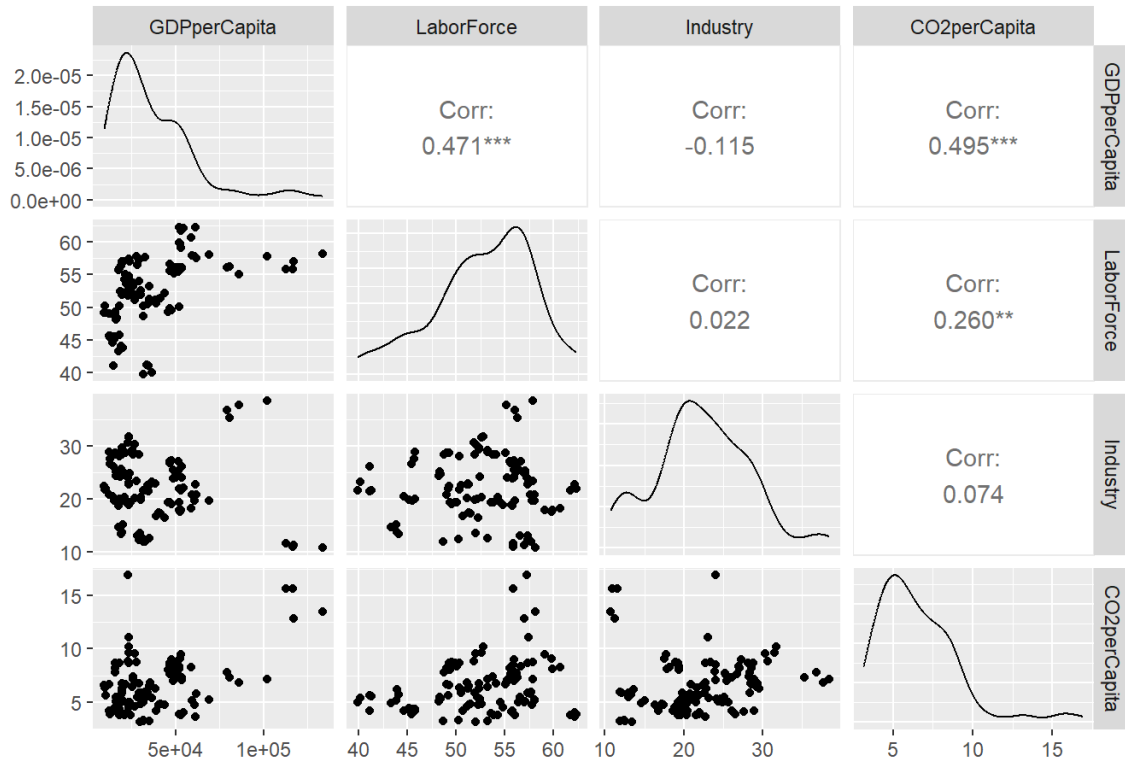


Figure 7: Scatterplot Matrix for EU

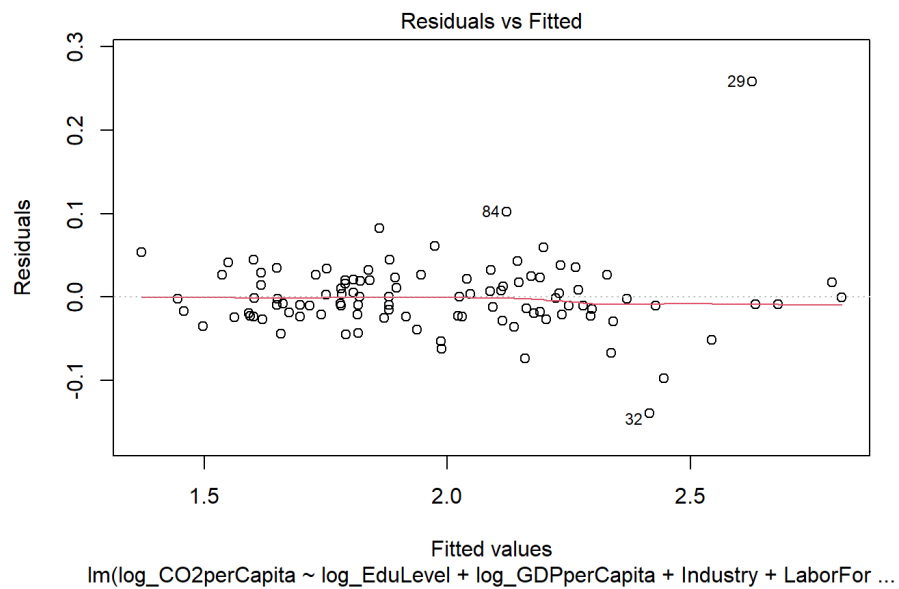
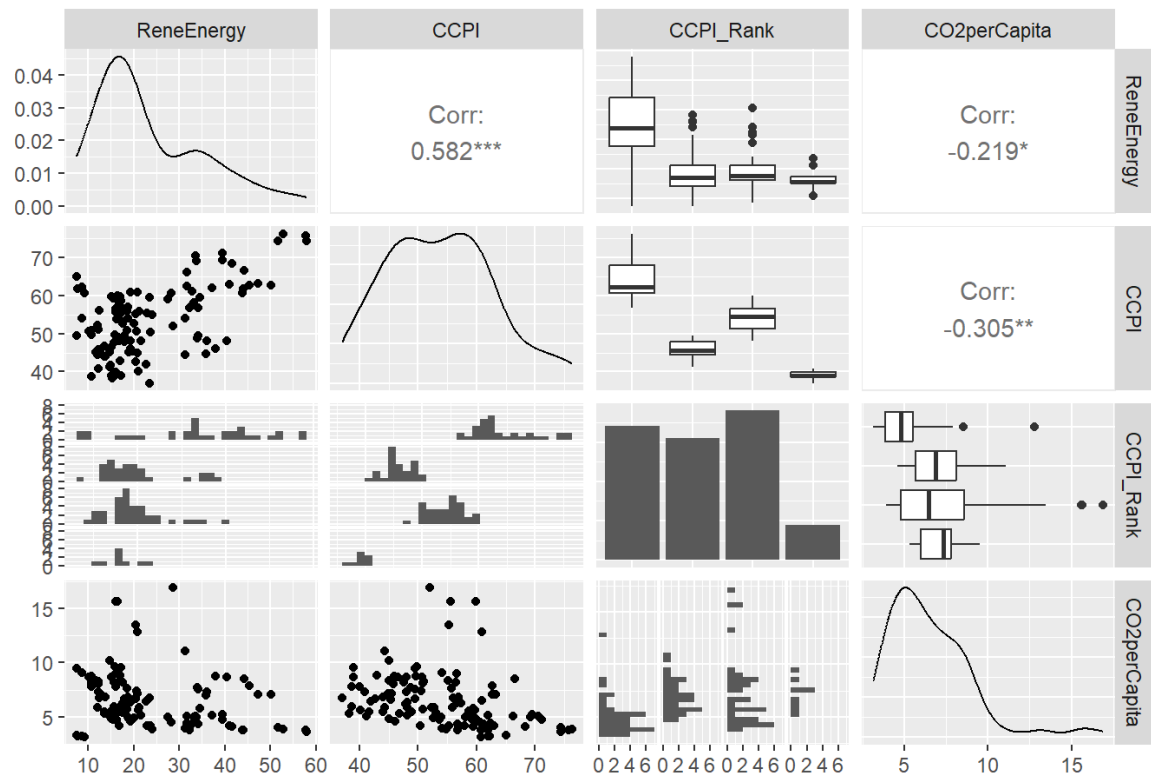


Figure 7: Residuals vs Fitted Plot of Panel Data Model

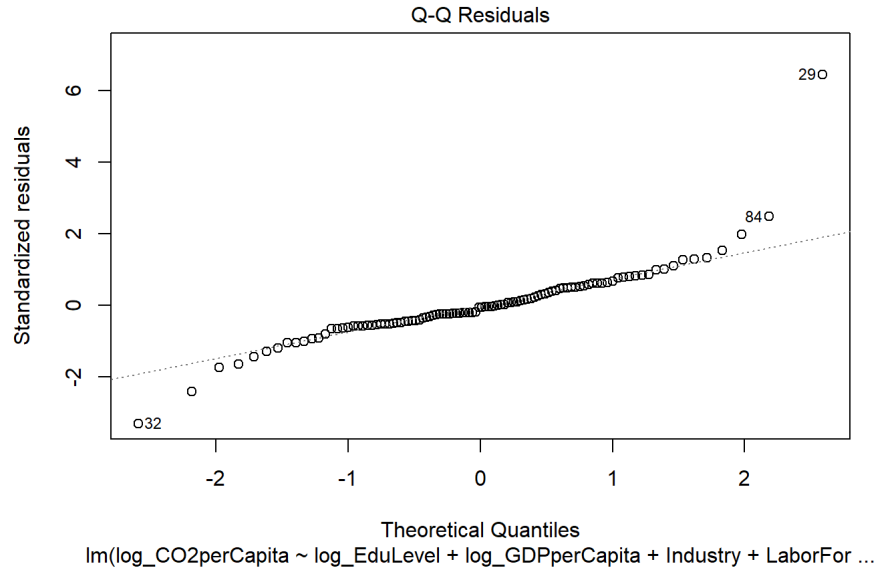


Figure 8: Q-Q Residuals Plot of Panel Data Model

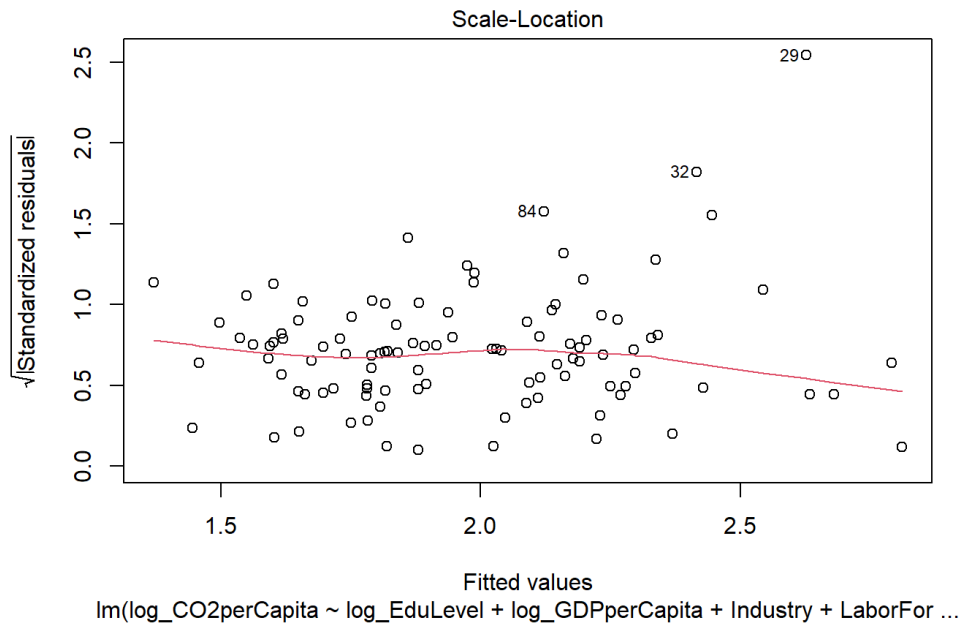
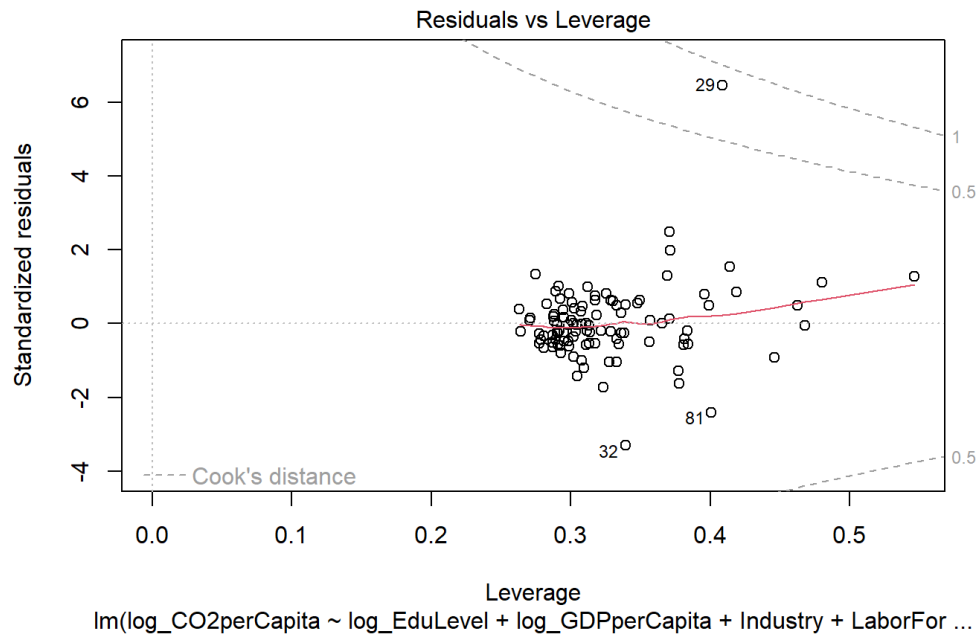


Figure 9: Scale-Location Plot of Panel Data Model



variable	Number of Observations	Minimum	Maximum	Median	Mean	Sd	25th Percentile	75th Percentile	Skewness	CountryGroup
CO2 Emissions Total (Mt CO2e)	24.0	8.1	744.6	121.2	221.4	235.3	66.6	322.4	1.2	Western Europe
CO2 Emissions per capita (t CO2e/capita)	24.0	4.2	15.6	8.2	8.7	3.0	7.5	9.0	0.9	Western Europe
Control of Corruption: Estimate	24.0	1.1	2.1	1.7	1.7	0.3	1.4	1.9	-0.1	Western Europe
Educational Attainment (Bachelor's or equivalent, population %)	24.0	14.8	41.7	30.1	28.7	8.6	20.7	35.6	-0.3	Western Europe
GDP per capita (current US\$)	24.0	39,179.7	133,711.8	50,747.0	60,410.2	27,800.8	46,722.5	53,162.9	1.7	Western Europe
Industry (% of GDP)	24.0	10.7	27.3	18.5	19.6	5.5	17.1	25.4	0.0	Western Europe
Labor Force Participation Rate (% of female population ages 15+)	24.0	49.3	60.6	55.9	54.9	3.5	51.4	56.7	-0.2	Western Europe
Renewable energy consumption (% of total final energy consumption)	24.0	7.4	36.0	16.1	17.9	8.5	12.1	19.0	1.2	Western Europe

Table 1: Summary Statistics for Key Variables for Western Europe

variable	Number of Observations	Minimum	Maximum	Median	Mean	Sd	25th Percentile	75th Percentile	Skewness	CountryGroup
CO2 Emissions Total (Mt CO2e)	44.0	7.2	330.1	35.4	63.4	86.1	14.3	76.6	2.3	Former Eastern Block
CO2 Emissions per capita (t CO2e/capita)	44.0	3.8	16.9	6.1	6.5	2.5	4.4	7.9	1.7	Former Eastern Block
Control of Corruption: Estimate	44.0	-0.3	1.6	0.5	0.4	0.5	0.0	0.7	0.6	Former Eastern Block
Educational Attainment (Bachelor's or equivalent, population %)	44.0	14.0	40.1	24.8	25.4	6.5	21.3	30.2	0.3	Former Eastern Block
GDP per capita (current US\$)	44.0	9,447.7	29,331.1	18,426.0	18,885.3	5,009.9	15,433.0	23,230.5	0.1	Former Eastern Block
Industry (% of GDP)	44.0	18.7	31.8	25.2	25.3	3.8	22.3	28.6	-0.2	Former Eastern Block
Labor Force Participation Rate (% of female population ages 15+)	44.0	41.1	57.8	52.1	51.5	4.3	48.6	55.2	-0.3	Former Eastern Block
Renewable energy consumption (% of total final energy consumption)	44.0	12.4	44.0	21.9	24.6	9.4	16.8	32.5	0.6	Former Eastern Block

Table 2: Summary Statistics for Key Variables for Former Eastern Block

variable	Number of Observations	Minimum	Maximum	Median	Mean	Sd	25th Percentile	75th Percentile	Skewness	CountryGroup
CO2 Emissions Total (Mt CO2e)	16.0	27.6	47.2	38.0	37.1	5.1	34.0	39.4	-0.1	Northern Europe
CO2 Emissions per capita (t CO2e/capita)	16.0	3.7	8.6	6.3	6.0	1.7	4.6	7.2	-0.1	Northern Europe
Control of Corruption: Estimate	16.0	1.5	2.3	2.1	2.0	0.3	2.0	2.2	-1.0	Northern Europe
Educational Attainment (Bachelor's or equivalent, population %)	16.0	25.1	37.7	32.4	31.8	3.6	30.3	33.5	-0.2	Northern Europe
GDP per capita (current US\$)	16.0	48,629.9	102,001.8	60,215.0	63,852.3	15,577.8	52,613.3	71,813.2	1.1	Northern Europe
Industry (% of GDP)	16.0	19.6	38.5	23.3	25.8	6.9	21.4	26.9	1.0	Northern Europe
Labor Force Participation Rate (% of female population ages 15+)	16.0	55.1	62.3	57.7	58.0	2.6	55.9	59.0	0.7	Northern Europe
Renewable energy consumption (% of total final energy consumption)	16.0	10.8	57.9	41.9	38.0	16.7	29.4	50.5	-0.6	Northern Europe

Table 3: Summary Statistics for Key Variables for Northern Europe

variable	Number of Observations	Minimum	Maximum	Median	Mean	Sd	25th Percentile	75th Percentile	Skewness	CountryGroup
CO2 Emissions Total (Mt CO2e)	24.0	1.6	340.3	51.7	113.1	127.3	7.2	237.9	0.7	Southern Europe
CO2 Emissions per capita (t CO2e/capita)	24.0	3.2	6.2	5.0	4.9	1.0	4.4	5.7	-0.6	Southern Europe
Control of Corruption: Estimate	24.0	-0.1	0.8	0.5	0.4	0.3	0.2	0.7	-0.4	Southern Europe
Educational Attainment (Bachelor's or equivalent, population %)	24.0	16.1	31.8	22.8	22.9	4.5	20.0	25.3	0.1	Southern Europe
GDP per capita (current US\$)	24.0	17,617.3	36,441.1	29,501.0	28,027.9	5,449.6	23,504.6	31,820.1	-0.4	Southern Europe
Industry (% of GDP)	24.0	11.9	23.2	17.0	16.8	3.9	13.0	20.1	0.1	Southern Europe
Labor Force Participation Rate (% of female population ages 15+)	24.0	39.8	57.7	51.9	49.8	6.0	43.9	54.0	-0.4	Southern Europe
Renewable energy consumption (% of total final energy consumption)	24.0	7.4	32.3	17.4	17.8	6.8	14.2	19.5	0.5	Southern Europe

Table 4: Summary Statistics for Key Variables for Southern Europe

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